Supplementary materials for: A Multi-Stream Temporal Context Model for narrative memory

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S1. Off-by-one correction in the reformatted recall table

The upstream `exp2.egg` file from[S1] encodes each recall event's provenance in a `temporal` HDF5 attribute that is 0-indexed against the `i{k}` subgroup names in the presented group. The initial reformat (`scripts/reformat_frfr_category.py`, commit `8684fa0`) treated `temporal` as if it were 1-indexed, which shifted every in-list recall's serial position down by one and silently re-coded the last word of every list as an extra-list intrusion. The corrected mapping (commit `405e272`, `sp = temporal + 1` for `temporal` $\in [0, 15]$, else `sp = 0`) restores the expected serial-position distribution (primacy at SP1–2, recency at SP14–16) and reduces the extra-list intrusion count from 399 to 10.

A regression test at `code/tests/test_frfr.py::test_recall_serial_position_shape` asserts $P(\text{recall}) > 0.3$ for the last three serial positions and an intrusion rate below 5%, so any future off-by-one would fail CI.

S2. Likelihood-surface flatness on small synthetic data

At FRFR-scale real data the MS-TCM likelihood surface is informative (grid scan over (β_G, w_G) gives a range of ~ 100 nats across the parameter simplex, vs a uniform-recall baseline ~ 1200 nats below the MLE). At the smaller scales used for quick sanity checks, the same grid scan shows an essentially flat surface (LL varies by ~ 2 nats around a uniform baseline of ~ -1064 nats). The practical consequence is that the parameter-recovery test at `code/tests/test_parameter_recovery.py` needs a synthetic dataset at FRFR scale (30 pt $\times$ 16 lists $\times$ 16 words $\times$ 4 categories $\approx$ 7,680 recalls) before the L-BFGS-B optimizer can reliably recover the true parameter vector; smaller datasets converge to whatever corner the initial point leans toward. This is a property of the likelihood surface, not a bug in the fitter.

S3. §4.4 rounding cascade

The model specification `notes/ms-tcm.pdf` §4.4 reports the bridge- condition composite similarity as 0.806 at $\beta_G = \beta_S = 0.5$, $w_G = 0.2$, $w_S = 0.8$, $m = 3$. This is a rounding-cascade artefact.

Using the closed form $\rho = \sqrt{1-\beta^2} = \sqrt{0.75}$ gives $\sigma(3) = 0.2 \cdot 0.75^2 + 0.8 \cdot \sqrt{0.75} = 0.1125 + 0.6928 = 0.80532\ldots$, which rounds to 0.805 at three decimals (not 0.806). The PDF's derivation used the truncated $\rho \approx 0.866$, then $\rho^4 \approx 0.563$, then rounded up at the final step. The unit test `test_section_4_4_numerical_anchor` asserts the closed-form values 0.86603 (grouped) and 0.80532 (bridge) at 10^{-6} tolerance. The agreement between the analytical closed-form and the model code is therefore exact; the apparent discrepancy with the PDF is purely a documentation issue.

Supplementary references

[S1] Jeremy R Manning, Emily C Whitaker, Paxton C Fitzpatrick, Madeline R Lee, Allison M Frantz, Bryan J Bollinger, Darya Romanova, Campbell E Field, and Andrew C Heusser. Feature and order manipulations in a free recall task affect memory for current and future lists, 2023. PsyArXiv preprint, `https://psyarxiv.com/erzfp`. Data and code: `https://github.com/ContextLab/FRFR-analyses`.